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Nonlinear effect of laser frequency modulation on molten pool width in deep penetration laser welding

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Abstract

This study investigates the nonlinear influence of laser modulation frequency on the molten pool width in deep penetration laser welding – a key concern in high-precision laser material processing. Through systematic experiments and numerical simulations, we examined the effect of modulation frequency on molten pool morphology under constant average laser power. Tests were conducted across a frequency spectrum from 20Hz to 20,000Hz, with representative cases at 20Hz, 200Hz, 2000Hz, and 20,000Hz to cover low, medium, and high-frequency domains. Special emphasis was placed on transition regions near 200Hz and 2000Hz, where nonlinear behaviors are most evident.

Outline

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A major finding of this work is the identification of a pronounced “U-shaped” variation in molten pool width over the 200–20,000Hz range: the width initially contracts significantly with increasing frequency, then gradually expands. This nonlinear contraction – expansion trend is explained by a shift in the thermo-fluid coupling mode within the molten pool across different frequencies. These insights offer a theoretical foundation and a practical frequency-window optimization strategy to improve process stability and geometric accuracy in modulated laser welding.

Keywords

Modulated laser; Laser deep melting; Molten pool width; Modulation frequencies

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1. Introduction

Laser deep penetration welding, characterized by its ultra-high energy density, sub-millimeter spatial precision, and non-contact nature, has become a cornerstone technology in high-end manufacturing sectors such as aerospace and new energy equipment. Moreover, laser technology has witnessed continuous development. Specifically, in the field of process optimization, machine learning and intelligent algorithms have been successfully applied to femtosecond laser drilling [1,2] and ultrashort pulse laser micromachining [3] processes, achieving adaptive optimization of processing parameters and enhanced quality stability. In the area of welding mechanisms and performance research, Zhang et al. [4] elucidated the formation mechanism of the waveform interface in Mg/Al dissimilar metal electromagnetic pulse welding from the perspective of molecular dynamics, providing a theoretical basis for defect control in dissimilar material joining. Xie et al. [5] systematically analyzed the influence of preweld conditions (hot isostatic pressing) on the strain age cracking susceptibility of the heat-

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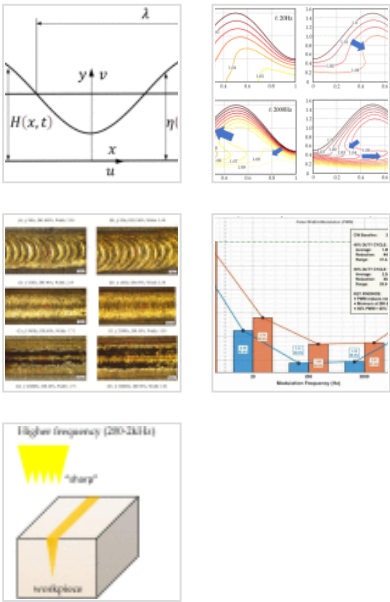
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Table 1. Parameters describing the molt...

affected zone in IN939 superalloy. M et al. [6], by comparing blue and infrared wavelengths such as 1030nm to 445nm, revealed the process applicability and energy coupling efficiency differences of different wavelengths in pure nickel welding. In the field of additive manufacturing, Wei et al. [7] constructed a multi-feature fusion deep learning model, achieving accurate identification of internal defects such as porosity and lack of fusion in laser additive components (average precision of 89.8%). V. N et al. [8], based on molten pool flow behavior analysis, systematically derived a computational model for the limiting rate in selective laser melting, clarifying the correlation mechanism between the rate and forming quality.

However, precise control of the molten pool width – a critical factor determining joint quality and geometric accuracy – still relies predominantly on two conventional approaches: adjusting laser power or varying the spot diameter. D. Y et al. [9] found that in current industrial practice, increasing laser power can widen the pool but significantly enlarges the heat-affected zone and induces undesirable deformation and grain coarsening. Conversely, reducing the spot diameter narrows the molten pool width but sacrifices penetration depth and is physically constrained by the diffraction limit of the optical system. Moreover, these methods are essentially static in nature and cannot meet real-time precision requirements in highly dynamic applications, such as curved surface processing or variable cross-section welding, the studies by Arkan et al., Li et al. and Ross et al. also confirm this limitation [[10], [11], [12]]. This paper systematically introduces, for the first time, laser modulation frequency as a novel methodological dimension for pool width control, addressing the inherent limitations of existing techniques.

As demonstrated by Tang et al. [13] and Gu et al. [14], modulated lasers, which convert continuous-wave beams into frequency-controlled pulse sequences, offer new degrees of freedom for dynamic molten pool control: the modulation frequency regulates the thermal pulse period, influencing the oscillatory response of the molten pool, while the duty cycle governs the energy deposition proportion within a single cycle. Nevertheless, Wu et al. and Kim et al. have pointed out that a clear understanding of how these laser parameters govern the internal multi-physics coupling in deep penetration welding is still

Table 2. . Mechanism analysis of droplet ...

lacking [15,16]. Key questions persist—for instance: How does high-frequency modulation suppress the transverse expansion of the molten pool? What correlation exists between the ultra-high-frequency “U-shaped” variation in molten pool width and keyhole instability? What constitutes the critical frequency criterion for thermo-fluidic mode transition? These knowledge gaps severely constrain the development of high-precision adaptive processing [17,18].

To address these challenges, this study proposes a research framework centered on the “frequency–pool width nonlinear mapping,” integrating systematic experimental investigation and numerical simulation: Multi-frequency experiments were conducted to systematically examine the frequency response of the molten pool width within the 20Hz–20,000Hz range (using TC4 titanium alloy); Cross-scale simulations were performed using an established thermal-flow-vapor coupled model to analyze the influence of modulation frequency on pool streamlines and temperature fields; Mechanism decoupling was achieved to reveal the hydrodynamic origins of the “contraction effect” (200Hz–2000Hz) and the “U-shaped reversal” ($f > 2000\text{Hz}$).

This work aims to quantitatively elucidate the nonlinear influence of modulation frequency on molten pool width, establish the corresponding numerical model, clarify the underlying physical mechanisms, and thereby provide a scientific basis for process optimization. The findings offer a theoretical foundation and a practical frequency-window optimization strategy, demonstrating significant potential for advancing high-precision laser deep penetration welding in industries requiring extreme consistency in weld formation, such as precision instrument manufacturing, automotive engineering, and aerospace component fabrication.

2. Longitudinal transport model in molten pools under modulated laser irradiation

2.1. Core mechanism governing longitudinal transport in molten pools under modulated laser irradiation

The impact of modulation frequency on deep penetration molten channels is not limited to the lateral morphology (molten width), but also profoundly affects the longitudinal energy transfer and channel penetration by regulating the molten pool's dynamic behavior. The core mechanism is manifested as:

2.1.1. Keyhole stability under periodic thermal loading

The essence of modulated laser irradiation is a periodic heat source ($Q_{\text{laser}}(t) = Q_0 \cdot g(f, t)$), each pulse triggers an instantaneous high-pressure vapor eruption at the keyhole tip (Back pressure $P_{\text{recoil}} \propto f^{1/2}$).

Low-frequency modulation (<200Hz): Long pulse cycles allow sufficient time for keyhole collapse and reformation. This amplifies fluctuations in longitudinal melt penetration depth and increases susceptibility to lack-of-fusion defects at the molten flow channels' root region.

Mid-to-high-frequency modulation (200–20,000 Hz): Short-cycle pulses maintain keyhole dynamic equilibrium. Sustained recoil pressure acts on the molten pool's base, driving molten metal axially downward along the penetration direction. This significantly enhances longitudinal energy transfer efficiency (thermal penetration depth $\delta_{th} \propto f^{-0.3}$).

2.1.2. Resonance effect of molten flow

The natural frequency f_{melt} of the molten pool is determined by melt viscosity μ , density ρ and characteristic dimension D :

$$f_{\text{melt}} \approx \frac{1}{2\pi} \sqrt{\frac{\sigma}{\rho D^3} - \frac{\mu}{\rho D^2}} \quad (1)$$

When the modulation frequency f approaches f_{melt} , the amplitude of fluid oscillations within the molten pool increases significantly [19,20]. This generates standing-wave flow

patterns, enhancing axial downward transport of molten metal and increasing melt penetration depth by 15–20%.

2.2. Coupling model of surface fluctuation and energy transfer in molten flow channels

To quantify longitudinal transport mechanisms, a surface fluctuation-energy transfer coupling model is established:

2.2.1. Dynamical equation of surface waves

The surface fluctuation amplitude $\eta(\mathbf{z}, t)$ evolves along the depth direction $\mathbf{z}$:

$$\frac{\partial^2 \eta}{\partial t^2} + \beta \frac{\partial \eta}{\partial t} = c^2 \frac{\partial^2 \eta}{\partial z^2} + \mathbf{F}(\mathbf{f}, t) \quad (2)$$

where:

$c = \sqrt{\sigma/(\rho h)}$ is the surface wave velocity. (σ : surface tension, h : melt layer thickness).

β : viscous damping coefficient.

$\mathbf{F}(\mathbf{f}, t)$: periodic forcing term driven by modulated laser irradiation.

2.2.2. Correlation equation of energy transfer

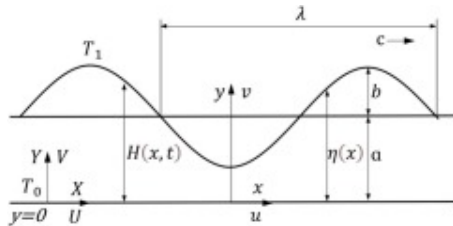
Surface wave amplitude η is positively correlated with longitudinal heat transport efficiency ξ :

$$\xi(z) = k_\eta \int_0^t \left| \frac{\partial \eta}{\partial z} \right| d\tau \quad (3)$$

where k_η is the fluctuation-heat transfer coupling coefficient, $\xi(z)$ higher indicates more efficient energy deposition at depth $\mathbf{z}$.

2.2.3. “Sawtooth effect” in frequency selection and physical analogy

As shown in Fig. 1, selecting modulation frequency f resembles sawtooth parameter optimization. Key molten pool parameters and their physical significance are summarized as shown in Table 1:



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Fig. 1. Geometric parameters of the melt channel during flow.

Table 1. Parameters describing the molten pool and their corresponding physical significance.

Formulation	Physical Interpretation
f_{melt} $\approx \frac{1}{2\pi} \sqrt{\frac{\sigma}{\rho D^3}} - \frac{\mu}{\rho D^2}$ $\propto \sqrt{\sigma/\rho D^3}$	Molten pool natural frequency dominated by surface tension. Frequency decreases with increasing characteristic dimension
$\xi(z) \propto \int \partial \eta / \partial z dt$	Longitudinal energy transfer efficiency scales with spatial gradient of surface waves. Higher gradients intensify melt mixing/convection.
$\lambda_{th} \propto f^{-1}$	Thermal perturbation wavelength inversely proportional to frequency. High frequencies promote energy localization.

The parameter correspondence is interpreted as follows:

Low-frequency regime (Coarse-tooth) ($f < 200\text{Hz}$): Excessively long thermal perturbation wavelength $\lambda_{th} \propto f^{-1}$ induces strong viscous attenuation of surface waves ($\eta \rightarrow 0$), resulting in weak longitudinal energy transfer. Consequently, melt penetration remains shallow and non-uniform.

Mid-frequency regime (Moderate-tooth) ($f \in 200\text{--}5000\text{Hz}$): Optimal match between thermal perturbation wavelength λ_{th} and molten flow channels characteristic dimension enhances surface wave resonance ($\eta \uparrow$). This resonance drives efficient downward melt convection, resulting in stable increase in penetration depth.

High-frequency regime (Fine-tooth) ($f > 5000\text{Hz}$): Excessively short thermal perturbation wavelength ($\lambda_{th} \ll D$) confines fluctuations to the superficial layer, preventing downward energy propagation. Thus, penetration depth exhibits saturation or even reduction.

2.3. Longitudinal transport model based on molten flow channels surface fluctuations

To clarify the problem, we simplify the solid–liquid phase change during molten flow channels formation, transverse flow effects, and recoil pressure within the keyhole as constants, focusing solely on frequency-dependent streamline behavior and we plot the graph of the resulting longitudinal wave speed, temperature distribution, and per-wavelength pressure rise by using a single surface wave profile of the molten flow channels [21,22].

First, we posit that the molten metal within the molten pool is modeled as a two-dimensional, unsteady, incompressible viscous flow. The model assumes the molten flow channels surface propagates in a wavelike pattern characterized by a wave width $2a$, while hypothetically maintaining its upper surface temperature remains fixed at T_1 .

The longitudinal transport characteristics of the molten flow are governed by a function. Then, a Cartesian coordinate system (X, Y) is established where wave propagation is parallel to the X-axis and the Y-axis lies orthogonal to the flow direction.

$H(X, t)$ is mathematically defined according to Fig. 1 as:

$$H(X, t) = a - b \cdot \cos \frac{2\pi(X - ct)}{\lambda} \quad (4)$$

where λ is wavelength; a is mean distance from the molten pool base to the mean wave position; b is amplitude; c is wave propagation speed along the X-direction.

The expressions are transformed using velocity relations:

$$x^* = X - ct, \quad y^* = Y, \quad u^* = U - c, \quad v^* = V \quad (5)$$

where U and V are determined by laser processing parameters [23]. And the governing equations in the coordinate system moving with the wave within the molten channel are as follows:

$$\frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0 \quad (6)$$

Expanding the above equations gives:

$$\begin{aligned} \rho \left(u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} \right) &= -\frac{\partial p^*}{\partial x^*} + \mu \left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}} \right) \\ \rho \left(u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} \right) &= -\frac{\partial p^*}{\partial y^*} + \mu \left(\frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\partial^2 v^*}{\partial y^{*2}} \right) \\ \rho c_p \left(u^* \frac{\partial T^*}{\partial x^*} + v^* \frac{\partial T^*}{\partial y^*} \right) &= \kappa^* \left(\frac{\partial^2 T^*}{\partial x^{*2}} + \frac{\partial^2 T^*}{\partial y^{*2}} \right) + Q_0 \end{aligned} \quad (7)$$

where Q_0 is the equivalent heat source parameter. By the transformation of the wave

framework, the function $\mathbf{H}(\mathbf{X},t)$ is expressed as:

$$\eta(\mathbf{x}^*) = a - b \cos\left(\frac{2\pi \mathbf{x}^*}{\lambda^*}\right) \quad (8)$$

The boundary conditions are given as follows:

At $\mathbf{y}^* = 0$:

$$\mathbf{v}^* = 0, \frac{\partial \mathbf{u}^*}{\partial \mathbf{y}^*} = 0, \frac{\partial \mathbf{T}^*}{\partial \mathbf{y}^*} = 0 \quad (9)$$

At $\mathbf{y}^* = \eta(\mathbf{x}^*)$:

$$\mathbf{u}^* = c, \mathbf{T}^* = \mathbf{T}_1, \mathbf{v}^* = \frac{2[\text{GSLPI}]b}{\lambda} \sin \frac{2\pi \mathbf{x}^*}{\lambda} \quad (10)$$

To simplify the problem formulation, we treat the volumetric flow rate $\mathbf{q}^*$ across the molten flow channels cross-section under laser irradiation as constant, which provides an auxiliary boundary condition:

$$\begin{cases} \psi^* = 0, (\mathbf{y}^* = 0) \\ \psi^* = \mathbf{q}^*, [\mathbf{y}^* = \eta(\mathbf{x}^*)] \end{cases} \quad (11)$$

where the relationship between volumetric flow rate of propagating waves in the molten channel $\mathbf{q}^*$ and time-averaged volumetric flow rate $\mathbf{Q}^*$ is governed by:

$$\mathbf{q}^* = \mathbf{Q}^* - c\mathbf{H} \quad (12)$$

To simplify and generalize the problem, the following dimensionless variables are introduced:

$$\mathbf{u} = \frac{\mathbf{u}^*}{c}, \mathbf{v} = \frac{\mathbf{v}^*}{c}, \mathbf{x} = \frac{\mathbf{x}^*}{\lambda}, \mathbf{y} = \frac{\mathbf{y}^*}{a}, \psi = \frac{\psi^*}{ca}, \mathbf{q} = \frac{\mathbf{q}^*}{ch}, \eta = \frac{\eta^*}{a}, \mathbf{p} = \frac{a^2}{\lambda \mu c} \mathbf{p}^* \quad (13)$$

By employing the velocity-stream function relationship $\mathbf{u} = \frac{\partial \psi}{\partial \mathbf{y}}$ 和 $\mathbf{v} = -\alpha \frac{\partial \psi}{\partial \mathbf{x}}$, the governing equations transform into the vorticity formulation as follows:

$$\alpha^2 \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega \quad (14)$$

And

$$Re \left(\frac{\partial \psi}{\partial y} \frac{\partial \omega}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \omega}{\partial y} \right) = \nabla^2 \omega, RePr \left(\frac{\partial \psi}{\partial y} \frac{\partial \theta}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \theta}{\partial y} \right) = \nabla^2 \theta + \beta \quad (15)$$

where Re is the Reynolds number, $Re = \frac{c\alpha\alpha}{v}$; Pr is the Prandtl number, $Pr = \frac{\mu c_p}{\kappa}$; α is the wave number, $\alpha = \frac{a}{\lambda}$; β is the heat generation parameter, and where:

$$\omega = \alpha \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}, \nabla^2 = \alpha^2 \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \quad (16)$$

Based on the above formulation, the boundary conditions for the stream function $\psi(\mathbf{x}, \mathbf{y})$ can be simplified as follows:

At $\mathbf{y}=0$,

$$\psi=0, \frac{\partial^2 \psi}{\partial y^2}=0, \frac{\partial \psi}{\partial x}=0, \frac{\partial \theta}{\partial y}=0 \quad (17)$$

At $\mathbf{y} = \eta(\mathbf{x})$,

$$\psi = q, \frac{\partial \psi}{\partial y} = -1, \theta = 1, \frac{\partial \psi}{\partial x} = 2\pi\phi \sin 2\pi x \quad (18)$$

3. Numerical analysis of longitudinal transport models for flow lines in keyhole molten pools during laser deep penetration

3.1. Numerical analysis of longitudinal transmission in the molten channel

Partial Differential Equation Toolbox (PDETool) enables numerical solution of heat conduction problems via the finite element method (FEM) without requiring custom programming. As established in prior sections, this work adopts a two-dimensional simplification for molten flow channel transport. The governing Eqs. (14), (15) are solved via FEM under boundary conditions (17), (18). When both Reynolds number (Re) and

wave number (α) approach zero, Eq. (15) reduces to an ODE. Considering molten flow continuity, the simulation processes a single waveform at each computational step, sequentially advancing to subsequent waveforms [24,25].

PDETool employs adaptive non-uniform meshing for parametric studies, achieving high convergence within 2–4 iterations. The model defines:

$$\psi = \sum_{k=1}^n \psi_k N_k, \theta = \sum_{k=1}^n \theta_k N_k, \omega = \sum_{k=1}^n \omega_k N_k \quad (19)$$

where the ψ_k and ω_k respectively denote the nodal approximations of ψ and ω at element vertices, and the system of equations is derived via the Galerkin finite element formulation as follows [26,27]:

$$\begin{cases} \int_{\Omega} \omega_1 \left(\alpha^2 \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \omega \right) d\Omega = 0 \\ \int_{\Omega} \omega_2 \left[Re \left(\frac{\partial \psi}{\partial y} \frac{\partial \omega}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \omega}{\partial y} \right) - \left(\alpha^2 \frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right) \right] d\Omega = 0 \\ \int_{\Omega} \omega_3 \left[RePr \left(\frac{\partial \psi}{\partial y} \frac{\partial \theta}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \theta}{\partial y} \right) - \left(\alpha^2 \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} \right) - \beta \right] d\Omega = 0 \end{cases} \quad (20)$$

where the ω_1 , ω_2 and ω_3 represent the weighting functions, and upon simplifying the aforementioned equations while incorporating Eq. (20) into Eq. (21), consideration of the discretized domain yields:

$$\begin{cases} \sum_i A_{ki}^e \psi_i - \sum_i B_{ki}^e \omega_i = S_n^{k^e} \\ \sum_i A_{ki}^e \omega_i + Re \sum_{i,j} C_{kij}^e \psi_i \omega_j = 0 \\ \sum_i A_{ki}^e \theta_i + RePr \sum_{i,j} C_{kij}^e \psi_i \theta_j = S_n^{k^e} + \beta S^{k^e} \end{cases} \quad (21)$$

where $A_{ki}^e = \int_{\Omega^e} \left(\alpha^2 \frac{\partial N_k}{\partial x} \frac{\partial N_i}{\partial x} + \frac{\partial N_k}{\partial y} \frac{\partial N_i}{\partial y} \right) d\Omega$, $C_{kij}^e = \int_{\Omega^e} \left(P_k \frac{\partial N_i}{\partial y} \frac{\partial N_j}{\partial x} - \frac{\partial N_j}{\partial x} \frac{\partial N_i}{\partial y} \right) d\Omega^e$,

$B_{ki}^e = \int_{\Omega^e} N_k N_i d\Omega$, $S^{k^e} = \int_{\Gamma} N_k d\Gamma$, $S_n^{k^e} = \int_{\Gamma} N_k \bar{S}_k d\Gamma$.

The global matrix equation is given by:

$$\mathbf{KA} = \mathbf{F} \quad (22)$$

The elements in the matrix are specified by Eq. (23):

$$\mathbf{K} = \{K_{ij}\} = \begin{bmatrix} -B_{ki}^e & A_{ki}^e & 0 \\ A_{ki}^e & Re C_{kij}^e \omega_i & 0 \\ 0 & Re Pr C_{kij}^e \psi_i & -A_{ki}^e \end{bmatrix}, \mathbf{A} = \{A_k\} = \begin{bmatrix} \omega_k \\ \psi_k \\ \theta_k \end{bmatrix}, \mathbf{F} = \{F_k\}$$

$$= \begin{bmatrix} S_n^{k^e} \\ 0 \\ S_n^{k^e} + \beta S^{k^e} \end{bmatrix} \quad (23)$$

Then the Newton-Raphson method is employed to solve the nonlinear equations, with the iterative procedure continuing until specified convergence criteria are satisfied [25,26,28].

Within the molten flow channel, the downward-acting laser-induced dynamic pressure loading and the upward recoil pressure are both reduced to streamline perturbations in the central region of surface waves. Furthermore, the pressure rise per wave cycle is governed by Eq. (24):

$$\Delta P_\lambda = \int_0^\lambda \frac{\partial p}{\partial x} dx \quad (24)$$

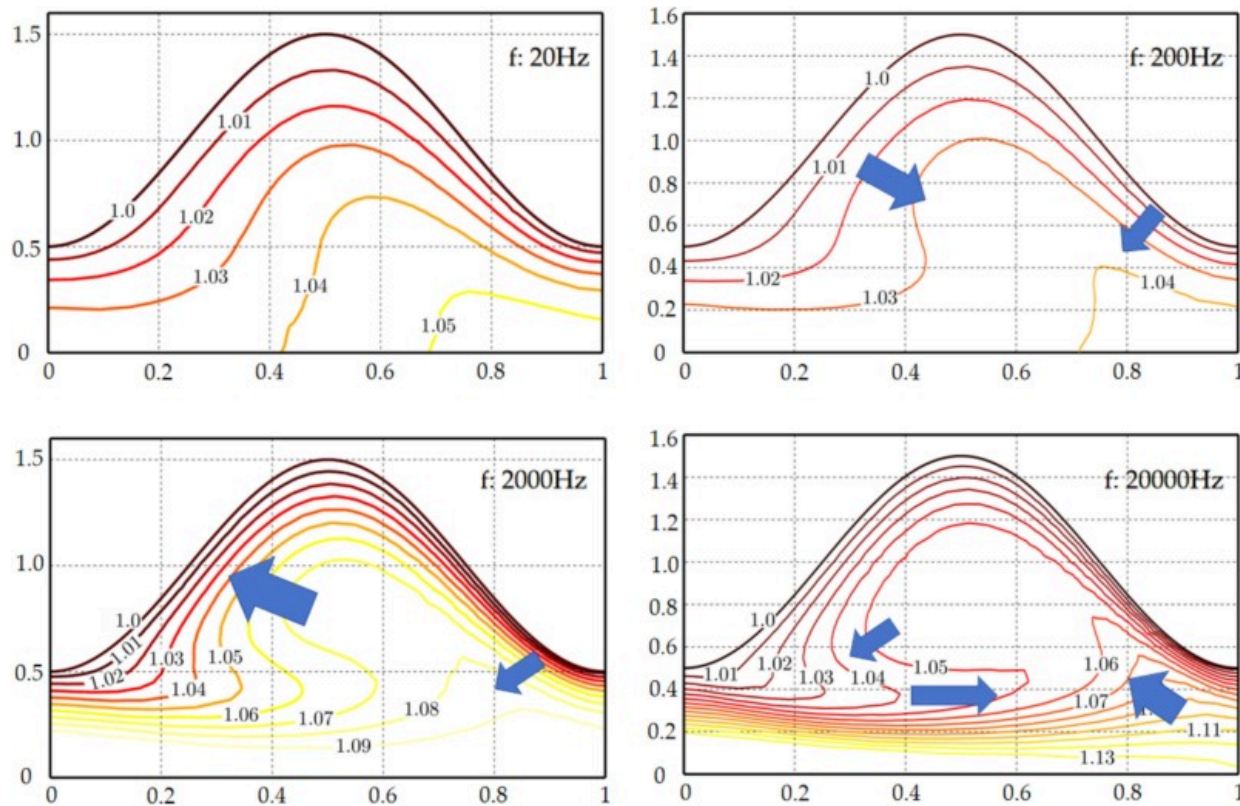
where the $\frac{\partial p}{\partial x}$ from the nondimensional form of the Navier-Stokes equations.

3.2. Modulated the internal melting flow transmission mechanism under the action of different frequencies of laser

Using PDETool toolbox, the molten pool region is discretized into a triangular element mesh, with each element comprising 6 nodes. First, a solution is obtained on each

individual triangular mesh. These elemental solutions are then assembled into a global system. Finally, the nonlinear system is solved via the Newton-Raphson method.

For comparative analysis, we selected a single representative waveform within the molten flow channel. The waveform is assumed to have identical width and height to exclusively analyze streamline motion patterns within the wave. Fig. 2 illustrates the simulated streamline distributions within the wave under different frequencies:

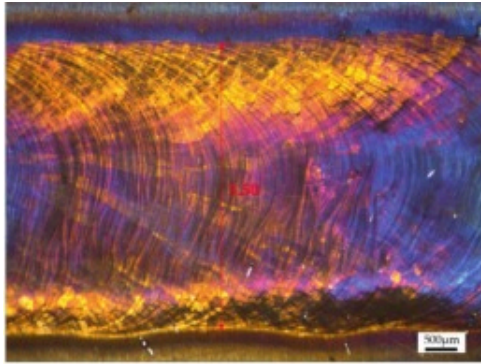
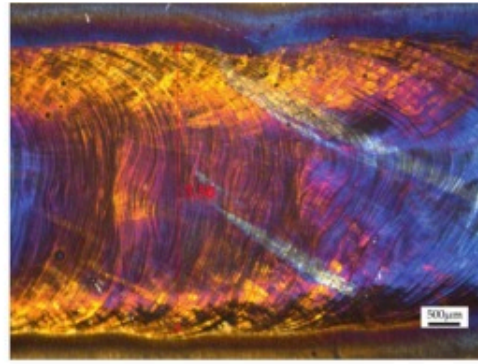
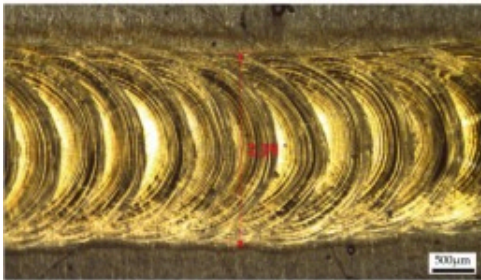
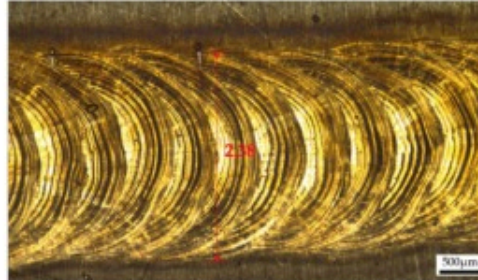
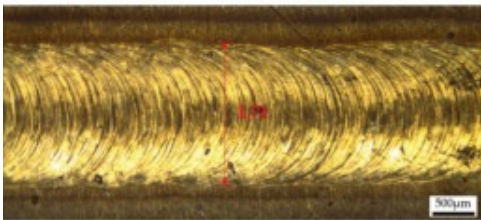
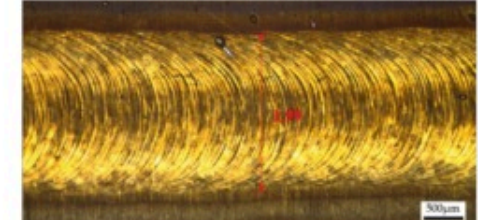
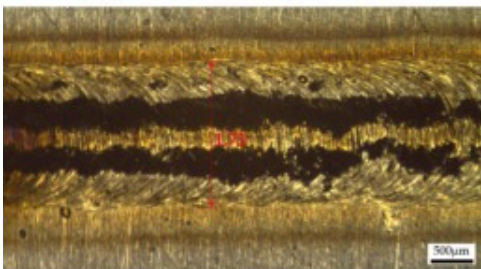
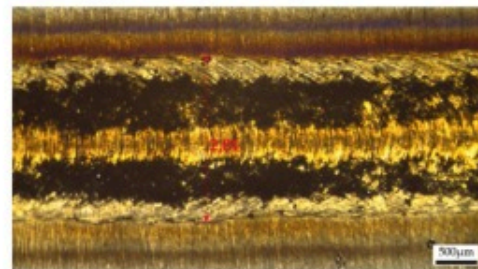


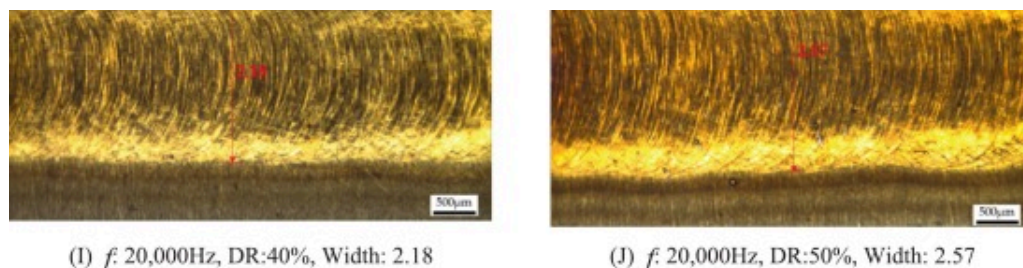
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Fig. 2. Streamline patterns of the melt channel at different frequencies.

Fig. 3 presents photographs of deep penetration surface weld beads under different laser parameters. It can be observed that at a modulation frequency of 0Hz (denoted as CW), corresponding to a 100% duty cycle, the molten pool width is 3.50mm (the unit is millimeters by default, consistent with conventional laser welding measurement standards). When the modulation frequencies are set to 20Hz, 200Hz, 2000Hz, and 20,000Hz (PWM), the obtained molten pool widths are as follows: at a 40% duty cycle, the widths are 2.19mm, 1.71mm, 1.73mm, and 2.18mm, respectively; at a 50% duty cycle, the widths are 2.38mm, 1.99mm, 2.01mm, and 2.57mm, respectively.

(A) f : 0Hz, DR:100%, Width: 3.50(B) f : 0Hz, DR:100%, Width: 3.50(C) f : 20Hz, DR:40%, Width: 2.19(D) f : 20Hz, DR:50%, Width: 2.38(E) f : 200Hz, DR:40%, Width: 1.71(F) f : 200Hz, DR:50%, Width: 1.99(G) f : 2000Hz, DR:40%, Width: 1.73(H) f : 2000Hz, DR:50%, Width: 2.01



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Fig. 3. Surface of single deep melting laser channel under different process parameters.

Furthermore, comparing the streamline distribution diagrams at these four frequencies in Fig. 2 reveals that as the frequency increases, the streamlines within the waves tend to transition from sparse to dense. This trend aligns with the observed increase in the number of weld layers with rising frequency in pulsed laser welding. Fig. 3C–J show surface images of the weld beads at different frequencies. At 20Hz, the spacing between the left and right spots reaches 25μm and is clearly visible in Fig. 3C and D. This observation is entirely consistent with the numerical analysis of streamlines at 20Hz in Fig. 2: the formation of waveforms within the molten pool is not solely determined by the single laser input frequency, but rather results from the coupling between the laser input frequency and the melt flow dynamics within the pool. This is manifested as the superimposition of secondary ripples caused by differences between the macroscopic flow waveform and microscopic flow velocities during molten flow transport. At this stage, the intermittent heating from low-frequency pulses causes the same area on the material surface to undergo multiple remelting cycles. When secondary waveforms with higher flow velocities within the longitudinally moving primary waveform successively break through the constraints of the upper-layer waveform, multiple discrete ripples form within the melt spot.

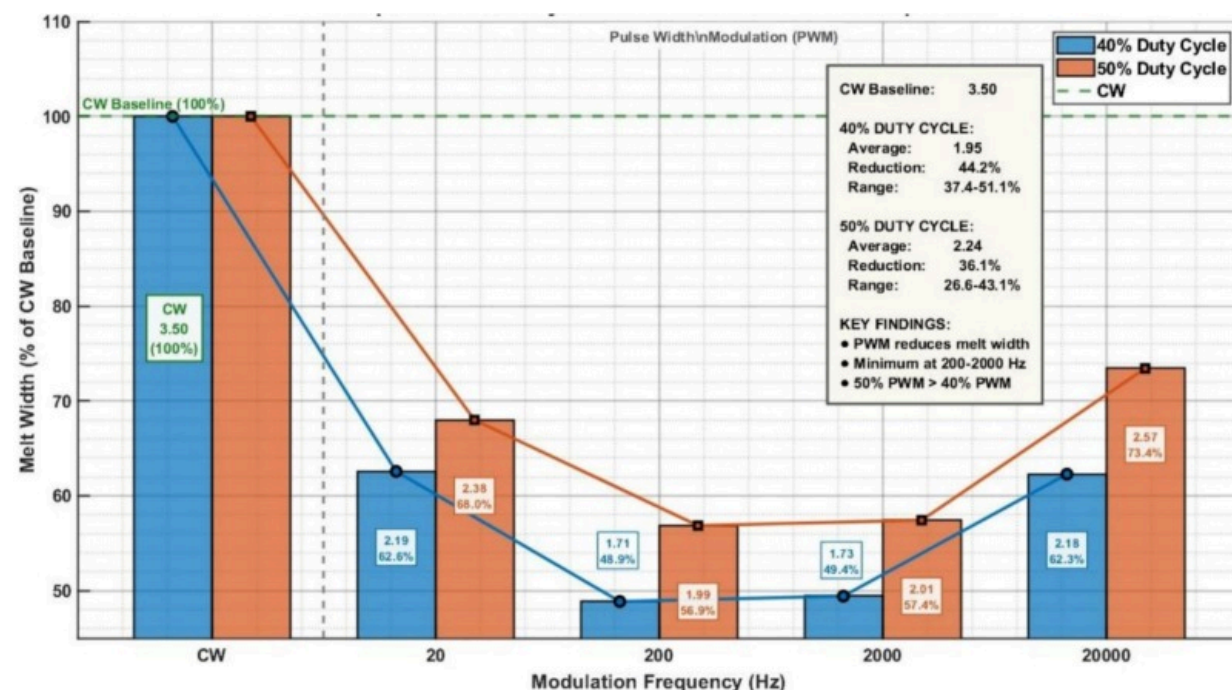
When the frequency is further increased to 200Hz, the overlap rate of the melt spots in the molten pool increases significantly, making it impossible to discern whether ripples

still exist within the spots. This phenomenon corresponds to the layered weld morphology characteristic of pulsed laser welding, where increased frequency leads to a greater number of weld layers and overlapping heat-affected zones from adjacent pulses. From the streamline diagrams in Fig. 2, it can be observed that the increased frequency elevates the velocity of intra-wave streamline formation by an order of magnitude compared to 20Hz. Combined with theoretical analysis of effective peak power density, the coupling effect between peak power density and pulse width becomes dominant at this stage: under the influence of liquid surface tension, internal high-velocity streamlines cannot rapidly break through the constraints of the preceding streamlines and instead fold back inward (as indicated by the arrows in the figure). This flow characteristic is consistent with the flow suppression effect dominated by viscous forces at high frequencies.

Upon further increasing the frequency to 2000Hz, an asymmetric streamline motion trend emerges (partly convex and partly concave), with the volume of the convex regions being rapidly replenished by the concave regions, resulting in highly pronounced depressions. This phenomenon can be explained by a two-dimensional transient model of rapid laser remelting: at low scanning speeds, depressions tend to form in the center of the molten pool with raised edges, and the interaction between low-frequency pulses exacerbates the asymmetry of solute flow within the pool. The variation trend in molten pool width is significantly greater than that in penetration depth, further amplifying the morphological distortion induced by streamline asymmetry.

When the frequency reaches 20,000Hz, the waveform would typically be expected to exhibit bilateral convex characteristics according to the pattern. However, Fig. 2 indicates the presence of flow tendencies in multiple directions. This is because the energy distribution of high-frequency pulsed lasers gradually approximates the Gaussian mode of continuous-wave lasers. At this stage, the remelting cycle effect weakens, and pool behavior approaches that of continuous-wave laser processing. Concurrently, high-frequency excitation enhances turbulence within the molten pool, leading to increased complexity in streamline directions. Experimental verification from corresponding

surface photographs of the weld beads shows that Fig. 3C, E, G, and I were obtained using process parameters with a 40% duty cycle, while Fig. 3D, F, H, and J used a 50% duty cycle. The difference between them solely reflects the level of energy per unit time (a higher duty cycle corresponds to greater effective energy input per unit time). To more clearly illustrate the variation pattern of the weld beads, the measured molten pool widths from ten experimental sets were analyzed and presented as curves in Fig. 4.



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Fig. 4. Comprehensive molten pool width in CW(unmodulated) vs modulated laser.

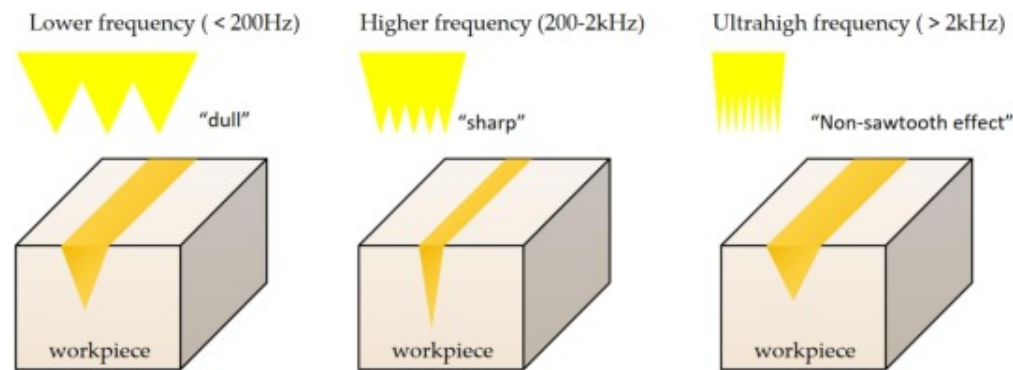
Comparing the surface molten pool widths of continuous-wave and modulated lasers in Fig. 4 reveals that the width for continuous-wave laser (3.50mm) is significantly greater than all widths for modulated lasers. This validates the proposed mechanism of molten pool width contraction under the intermittent energy action of modulated lasers in this

study. The core of this mechanism lies in the periodic energy on–off cycling of modulated lasers, which subjects the molten pool to “rapid heating-instantaneous cooling” cycles. Compared to the continuous energy input of CW lasers, this significantly reduces heat accumulation and the driving force for lateral diffusion, while simultaneously narrowing the heat-affected zone. From an energy input perspective, a higher duty cycle represents greater energy per unit time, thus resulting in a larger molten pool width. This aligns with the general rule in pulsed laser welding that “energy density is positively correlated with molten pool width.” It is noteworthy that the molten pool width is larger at low (20Hz) or high (20,000Hz) frequencies, while the widths at 200Hz and 2000Hz are relatively similar and smaller. It is inferred that this range constitutes the deep penetration molten pool width contraction zone. This conclusion is highly consistent with the streamline simulation results, indicating that within this frequency range, the flow of streamlines within the weld bead exhibits a significant contraction trend, associated with the dual effects of flow suppression at high frequencies and remelting cycles at low frequencies.

Overall, increasing energy leads to a corresponding increase in molten pool width. However, with increasing modulation frequency, the molten pool width exhibits a non-monotonic trend, initially decreasing and then increasing. This pattern reflects the dual regulation of frequency on energy absorption and flow behavior within the molten pool: in the low-frequency range (20Hz), the single-pulse energy is relatively high, allowing sufficient time for the pool to spread; in the mid-frequency range (200–2000Hz), the frequency matches the natural oscillation frequency of the pool, resulting in the highest energy utilization efficiency and the strongest flow suppression effect; in the high-frequency range (20,000Hz), the modulation mode approximates that of a continuous-wave laser, and enhanced turbulence causes the molten pool width to increase again. Analysis combined with the numerical calculation results in [Fig. 2](#) shows that the simulated streamline results are in good agreement with the experimental measurements, further confirming the reliability of this process parameter control mechanism.

4. Mechanisms of molten flow transport within channels under modulated laser frequencies

As illustrated in Fig. 5, the modulated laser beam exhibits a characteristic frequency-dependent “sawtooth effect” during deep penetration processing. The underlying physical mechanism involves the competitive matching between the thermal perturbation wavelength and the molten pool characteristic dimension:



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Fig. 5. Longitudinal action of the modulated laser at different frequencies.

In the low-frequency coarse-tooth mode ($f < 200\text{Hz}$), the molten pool response is dominated by large-scale Marangoni vortices, which govern transverse thermal transport. Physically, this mode corresponds to the action of a coarse-tooth saw blade (tooth pitch $> 3\text{ mm}$), resulting in wide and rough cutting paths.

A critical transition occurs in the intermediate-frequency sharp-tooth mode ($200\text{Hz} \leq f \leq 20,000\text{Hz}$), where the thermocapillary suppression effect is activated. Consequently, the molten pool width logarithmically contracts with increasing frequency. The physical analogy here is a fine-tooth saw blade (tooth pitch $0.05\text{--}0.5\text{ mm}$), enabling high-precision, narrow-kerf processing.

Beyond this range, in the high-frequency quasi-continuous mode ($f > 20,000\text{Hz}$), the mechanism governing the sawtooth effect reverses. Micro-turbulence is reactivated, enhancing edge thermal diffusion and leading to a recovery molten pool width. This final mode is analogous to an ultra-fine-tooth blade (tooth pitch $< 0.02\text{mm}$) inducing surface fluidization, equivalent to continuous-wave (CW) laser action.

However, in actual deep penetration processing, the modulated laser interaction cannot be simply reduced to a sawtooth analogy, as the energy-matter interaction is exceedingly complex. During modulated laser deep penetration, the spatiotemporally concentrated energy input of the high-energy laser beam primarily acts on the keyhole region at the front of the molten pool. This creates extremely high temperature gradients in this zone, inducing intense thermocapillary forces (Marangoni forces). Driven by these forces, molten metal in the superficial layers at the front and center of the molten pool forms high-speed thermocapillary convection, flowing outward towards the molten pool edges (regions of lower surface tension). Simultaneously, within the central region of the molten pool, the intense heat input and significant melt temperature gradients also drive molten flow penetrating downwards, forming deep convective vortex structures directed towards the molten pool bottom. This complex multi-directional flow structure, which comprises both the superficial outward flow and the deep downward-penetrating flow, represents the key hydrodynamic factor governing the morphology of the molten pool, particularly the molten pool width.

Significantly, the thermo-fluid coupling behavior within the molten pool is inherently transient, being strongly modulated by droplet transfer (if present) and the intense dynamic behavior of the keyhole. As the core region for energy absorption and vapor recoil pressure, the keyhole exhibits stability and oscillation characteristics that undergo significant alterations with changes in the modulated laser pulse frequency. When pulsed laser irradiation at different frequencies interacts with the metal, each pulse triggers a rapid keyhole opening phase (dominated by vaporization pressure), followed by instability, contraction, and eventual collapse (governed by surface tension and hydrodynamic forces). This process maintains the keyhole walls in a state of continuous,

nonlinear, real-time fluctuation. Such periodic oscillations of the keyhole not only directly impact the energy absorption efficiency at the front of the molten pool, but also strongly perturb the internal temperature field and velocity field of the molten pool. Consequently, they dynamically alter both the magnitude and direction of the thermocapillary forces, ultimately feeding back into the macroscopic flow patterns of the molten pool. This real-time, strongly coupled interaction mode – involving the thermophysical state (temperature field) and hydrodynamic state (velocity field, pressure field) of the molten pool – constitutes the fundamental physical mechanism responsible for the complex nonlinear variations in molten pool width with modulation frequency, such as contraction within specific intervals and the observed “U-shaped” molten pool width.

In modulated laser deep penetration processing, the relative distance between the molten metal droplets impact location and the keyhole is a critical parameter for regulating molten pool dynamics. When molten metal droplets generated at specific frequencies impact the vicinity of the keyhole rear wall in a spatially and phase-synchronized manner (distance threshold $< 200\mu\text{m}$), they generate a transient impact pressure as high as 10^4 – 10^5 N/m^2 . This impact force creates an asymmetric pressure field at the top of the molten pool, driving the molten metal to form a downward-forward composite jet. This jet exerts a dual effect:

1. It accelerates the forward flow velocity of the melt along the keyhole rear wall (increasing by 50%–80% compared to the baseline state). The high-speed molten flow compresses the rear neck of the keyhole, leading to a localized necking effect (keyhole width reduction of 15%–30%).
2. It induces severe oscillations of the keyhole wall (amplitude reaching 50–100 μm), disrupting the stability of the gas–liquid interface at the keyhole boundary.

These dynamic disturbances influence energy transmission through two primary pathways:

1. Distortion of laser energy density distribution: The oscillating keyhole wall generates a dynamic Fresnel absorption effect, causing the laser irradiation zone to exhibit alternating bands of high and low energy density (fluctuation amplitude > 30%).
2. Imbalance of vapor recoil pressure: A supercritical ablation pressure (> 500kPa) forms within the laser direct irradiation zone, while the pressure in the shadow zone drops sharply to < 100kPa. The resulting pressure gradient force accelerates keyhole distortion.

The aforementioned coupled effects trigger a self-reinforcing feedback loop:

Molten Metal Droplet Impact → Keyhole Deformation → Non-uniform Energy Distribution → Ablation Pressure Imbalance → Enhanced Keyhole Instability → Increased Closure Probability.

When the molten metal droplet impact location shifts outward to a safe distance (> 500 μ m), the impact pressure attenuates to < 10³ N/m², and the amplitude of keyhole wall oscillations decreases below 20 μ m. Consequently, the keyhole enters a quasi-steady state, and its stability index (defined as open time/pulse period) increases to above 0.8. The mechanisms governing molten metal droplet impact effects under different modulation frequencies are summarized in [Table 2](#).

Table 2. . Mechanism analysis of droplet impact effects under different modulation frequencies.

Frequency Range	Molten Metal Droplet Impact Location Effect	Keyhole Response	Molten Pool Width Variation
0–200Hz	Spatial dispersion of molten metal droplet	Reduced impact probability → Weakened necking	Dominant shrinkage trend

Frequency Range	Molten Metal Droplet Impact Location Effect	Keyhole Response	Molten Pool Width Variation
		effect	
200–2000Hz	30% reduction in molten metal droplet diameter	Increased impact energy density but decreased penetration depth	U-shaped transition zone
2000–20000Hz	Spatio-temporal resonance between molten metal droplet impact and laser pulses	Synergistic interaction of vapor recoil and impact forces	Minimum molten pool width
>20000Hz	Melt inertia suppresses droplet formation	Impact energy dissipated by molten pool turbulence	Molten pool width recovery

Under specific laser modulation frequencies, the penetrative flow of molten metal from the molten pool surface toward the bottom is enhanced, significantly improving heat transfer efficiency between the upper and deep regions of the molten pool. This process promotes increased thermal energy transfer toward the bottom of the molten pool. Consequently, the temperature gradient at the pool surface decreases, leading to a corresponding increase in surface tension coefficient. This effectively suppresses material accumulation at the leading edge of the molten pool, thereby improving uniformity of mass transport within the liquid phase.

Simultaneously, the dynamic impact effect generated during molten metal droplet transfer into the molten pool disturbs keyhole stability and alters local flow structures. This impact influences molten pool dynamics through two mechanisms:

1. Direct disruption of force equilibrium on keyhole walls, inducing keyhole necking or oscillations;

2. Generation of pressure fluctuations within the molten pool, restructuring melt convection paths.

In the trailing region of the molten pool, molten flow dominated by the Marangoni effect exhibits a downward and rear-upward composite flow pattern. When this flow reaches the solidification interface (solid–liquid two-phase zone), it encounters physical obstruction from growing dendritic structures, forcing a flow reversal that forms an upward backflow. Under the coupled action of thermocapillary forces and solidification resistance, this redirected melt evolves into a secondary vortex flow propagating along the solidification front.

The dynamic adjustment of these thermo-fluid coupling mechanisms – enhanced melt penetration improving heat conduction, molten metal droplet impact disturbing flow fields, and solidification front restructuring vortex flows – exhibits strong frequency dependence:

1. In the 200–20,000Hz range, high-frequency pulses shorten melt response time, suppressing metal accumulation at the molten pool leading edge and causing molten pool width contraction.
2. Above a critical frequency ($>20,000\text{Hz}$), enhanced inertial effects of the molten material combined with diminished obstruction to molten flow by the solidification front thereby reduce secondary vortex intensity. Concurrently, vapor recoil pressure dominates keyhole behavior. These factors collectively drive a “U-shaped” evolution of molten pool width.

5. Conclusion

This paper pioneers a novel method for directionally manipulating deep-penetration molten pool morphology by modulating laser frequency. Through synergistic validation

using multi-scale mathematical models and high-resolution experiments, its physical mechanism is systematically revealed:

1. A novel sawtooth effect mechanism model was developed, introducing a matching theory between thermal disturbance wavelength and molten pool characteristic dimension. This led to a universal mathematical equation. Critically, numerical analysis demonstrates that laser modulation frequency (f) governs the competition between thermocapillary (Marangoni) force and recoil pressure by modulating the alignment of thermal disturbance wavelength with molten pool characteristic dimension. A key finding establishes the criterion for transverse molten flow suppression: thermocapillary response time must match the modulation period. Specifically, the optimal suppression occurs within the frequency range of 200–2000Hz, where the molten pool width reaches its minimum values of 1.71–1.73mm (40% duty cycle) and 1.99–2.01mm (50% duty cycle), representing a 48.9–50.6% reduction compared to continuous wave processing.
2. Through integrated experimentation and modeling, we established a complete quantitative relationship between laser modulation frequency and molten flow transport for the first time. This breakthrough deciphers the hydrodynamic mechanism governing U-shaped molten pool width formation. Our longitudinal melt channel transport model validated by simulations and experiments reveals how frequency modulation regulates molten flow, thereby solving the central challenge of controllable melt transport in deep-penetration processing. The experimental data demonstrate a distinct non-monotonic trend: at low frequency (20Hz), the molten pool width measures 2.19mm (40% duty cycle) and 2.38mm (50% duty cycle); while at high frequency (20,000Hz), it increases to 2.18mm and 2.57mm respectively, confirming the existence of an optimal frequency window for molten pool width control.

3. The identified molten pool width contraction mechanism operates through dual physical effects: at medium frequencies (200–2000Hz), the thermal disturbance wavelength matches the molten pool characteristic dimension, maximizing flow suppression efficiency; while at low frequencies (20Hz), discrete remelting cycles with 25 μ m spot spacing dominate, and at high frequencies (20,000Hz), turbulence enhancement becomes predominant. This fundamental understanding provides a scientific basis for precise regulation of welding morphology, achieving targeted control of molten pool width within 1.71–2.57mm across different processing parameters.

The agreement between numerical simulations and experimental measurements confirms the reliability of the proposed mechanism. This research establishes a universal physical model for laser material processing optimization, offering theoretical guidance for industrial applications requiring precise energy distribution control.

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Not applicable.

Informed Consent Statement

Not applicable.

CRediT authorship contribution statement

Shen Long-Zhang: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Chen Wen-Liang:** Visualization, Validation. **Wu Xiao-Yan:** Visualization, Validation, Supervision. **Qiong Liu:** Visualization, Supervision. **Qiu Changjun:** Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data generated or analyzed during this study are included in this article.

References

- [1] C. Wang, Z. Zhang, X. Jing, Z. Yang, W. Xu
Optimization of multistage femtosecond laser drilling process using machine learning coupled with molecular dynamics
Opt. Laser Technol., 156 (108442) (2022)
[Google Scholar ↗](#)
- [2] Z. Zhang, S. Liu, Y. Zhang, C. Wang, S. Zhang, *et al.*
Optimization of low-power femtosecond laser trepan drilling by machine learning and a high-throughput multi-objective genetic algorithm
Opt. Laser Technol., 148 (107688) (2022)
[Google Scholar ↗](#)
- [3] Z. Zhang, Z. Yang, C. Wang, W. Xu
Accelerating ultrashort pulse laser micromachining process comprehensive optimization using a machine learning cycle design strategy integrated with a physical model

J. Intell. Manuf., 35 (2024), pp. 449-465

[Crossref ↗](#) [Google Scholar ↗](#)

- [4] Z. Long, H. Yongliang, D. Xuemei, L. Xin, Y. Limeng, *et al.*
Formation mechanism of waveform interface in Mg/Al electromagnetic pulse welding

Mater. Today Commun., 45 (2025), Article 112401



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [5] J. Xie, C. Geng, J. Song, Y. Chen, S. Wang, P. He
Effect of preweld conditions on the strain ageing cracking susceptibility of IN939 superalloys heat affected zone

J. Mater. Res. Technol., 36 (2025), pp. 5684-5695



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [6] M. Mbus, R. Pordzik, A. Krmer, T. Mattulat
Process comparison of laser deep penetration welding in pure nickel using blue and infrared wavelengths

Welding World: J. Int. Inst., 6 (2024), p. 68

[Google Scholar ↗](#)

- [7] W. Yang, X. Gan, J. He
Defect identification of 316l stainless steel in selective laser melting process based on deep learning

Processes, 12 (6) (2024)

[Google Scholar ↗](#)

- [8] V.N. Chuvildeev, A.V. Semenycheva, S.V. Shotin, M.Y. Gryaznov
Approaches to determining the limiting rate of selective laser melting of metals and alloys

Phys. Met. Metall., 125 (9) (2024), pp. 936-947

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [9] D.Y. Pimenov, L.F. Berti, G. Pintaude, G.X. Peres, Y. Chaurasia, N. Khanna, *et al.*
Influence of selective laser melting process parameters on the surface integrity of difficult-to-cut alloys: comprehensive review and future prospects
Int. J. Adv. Manuf. Technol., 127 (3–4) (2023), pp. 1071-1102
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [10] Z. Ekrem, H. Arkan
Investigation of mechanical properties of laser welded dual-phase steels at macro and micro levels
Opt. Laser Technol., 157 (2023)
[Google Scholar ↗](#)
- [11] W. Tingting, P. Li, X. Yu, *et al.*
High-energy hundred-picosecond fiber-solid hybrid laser and its application in laser-induced damage in PIN photodiode
Laser Phys., 30 (3) (2020), Article 036004
[Google Scholar ↗](#)
- [12] C. Ross, Z. Cang, P. Niranjana, *et al.*
Keyhole threshold and morphology in laser melting revealed by ultrahigh-speed x-ray imaging
Science, 363 (6429) (2019), p. 849
[Google Scholar ↗](#)
- [13] C. Tang, J.L. Tan, C.H. Wong
A numerical investigation on the physical mechanisms of single track defects in selective laser melting
Int. J. Heat Mass Transf., 126 (2018), pp. 957-968
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] Gu. Dongdong, M. Xia, D. Dai

On the role of powder flow behavior in fluid thermodynamics and laser processability of Ni-based composites by selective laser melting

Int. J. Mach. Tool Manuf., 137 (2019), pp. 67-78

[Google Scholar ↗](#)

- [15] Y.-C. Wu, C.-H. San, C.-H. Chang, *et al.*
Numerical modeling of melt-pool behavior in selective laser melting with random powder distribution and experimental validation

J. Mater. Process. Technol., 254 (2018), pp. 72-78



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [16] K. Byunggi, I. Ryoichi, K. Syunya, *et al.*
Effect of beam profile on nanosecond laser drilling of 4H-SiC

J. Laser Appl., 30 (3) (2018), Article 032207

[Google Scholar ↗](#)

- [17] Y. Bai, Y. Yang, Z. Xiao, *et al.*
Process optimization and mechanical property evolution of AlSiMg0.75 by selective laser melting

Mater. Des., 140 (2018), pp. 257-266



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [18] A.S. Raihan, H. Khosravi, T.H. Bhuiyan, I. Ahmed
An augmented surprise-guided sequential learning framework for predicting the molten pool geometry

J. Manuf. Syst., 75 (2024), pp. 56-77



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [19] M. Mbus, R. Pordzik, T. Seefeld
Influence of intensity distributions on the process dynamics during laser deep penetration welding of pure nickel with a flexible ring mode laser

SOURCE

J. Laser Appl., 36 (4) (2024), Article 042073

[Google Scholar ↗](#)

- [20] R. Pordzik, T. Seefeld
Identifying the direction of weld path deviations in laser deep penetration welding of hidden t-joints by means of oct

J. Laser Appl., 36 (3) (2024), Article 032031

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [21] Y. Li, Y. Gao, H. Pan, D. Tao, H. Zhang
Keyhole depth prediction in laser deep penetration welding of titanium alloy based on spectral information

Metals (2075-4701), 15 (5) (2025)

[Google Scholar ↗](#)

- [22] H. Zhang, L. Yang, Y. Huang
A study on the dynamic behavior of plasma plume in laser deep penetration welding of a304 stainless steel based on three-probe detection

Measurement, 250 (2025)

[Google Scholar ↗](#)

- [23] M. Yang, D. Cai, S. Gao, W. Fang, Z. Luo, J. Huang, *et al.*
Research on the behavior of molten pool in laser deep-penetration welding

J. Laser Appl., 36 (4) (2024), Article 042049

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [24] C. Jia, Y. Huang, S. Zhao, C. Li, J. Yuan, F. Zhang, *et al.*

Statistical analysis of plasma thermodynamic behavior based on laser deep penetration welding of tc4 titanium alloy

Measurement, 225 (2024), p. 9

[Google Scholar ↗](#)

- [25] J. Wahl, C. Frey, J. Powell, F. Zai, M. Haas, S. Olschok, *et al.*
Laser-plume interactions in deep-penetration remote laser welding of stainless steel
Opt. Laser Technol., 186 (2025)
[Google Scholar ↗](#)
- [26] G. He, X. Gao, L. Li, P. Gao
Oct monitoring data processing method of laser deep penetration welding based on hdbscan
Opt. Laser Technol., 179 (2024)
[Google Scholar ↗](#)
- [27] D. Brinkmeier, J. Wagner, M. Sawannia, J. Wahl, T. Menold, P. Stritt, *et al.*
Influence of the laser-plume interaction on the beam profile during deep-penetration welding with coherent beam shaping
Opt. Express, 33 (2) (2025), pp. 2702-2711
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [28] Y. Hu, C. Wang, Z. Liu, Z. Zhao, F. Yan
The formation mechanism and process regulation of collapse defects in high-power laser penetration welding of 20mm thick 316l stainless steel plates
Int. J. Heat Mass Transf., 247 (2025)
[Google Scholar ↗](#)

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